Agile Sprint Plan & Iteration Report: V1 to V2

MQ Sentinel: Autonomous Satellite Space Debris Removal (PoC)

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1. Executive Summary & Sprint Goal

Project: MQ Sentinel
Objective: Build a proof-of-concept autonomous satellite capable of locating and removing simulated space debris (represented by 38 kHz infrared beacons) using an eye-safe laser pointer. The system must operate fully contained within a 170 mm x 150 mm PVC pipe, utilizing a DC gearmotor reaction wheel for attitude control while suspended from a low-friction swivel.

Sprint Goal: Transition from the V1 proof-of-concept (subsystem integration demonstration) to a fully viable V2 prototype. Execute an impeccable, standard-paced sprint that flawlessly closes the loop on V1 failures (specifically IR signal bounce) using a highly disciplined product backlog and objective tracking methodologies.

2. Scrum Team & Roles

To ensure strict backlog discipline and accountability, the team is structured with clear delineations between hardware, software, and Agile management:

Name	Role	Core Responsibilities
Seth	Product Owner / Hardware	Backlog prioritization, stakeholder alignment, circuit design and wiring, hardware design and component integration .
Gabriel	Scrum Master / Software	Process enforcement, objective tracking, removing blockers, software implementation, telemetry recording and processing, software architecture.
Harvey	Software Developer	Software development, input handling, input processing.
Richard	Software Developer	software development and testing, focused on motor actuation and calibration.
Allan	Hardware Engineer	Physical structural assembly, 3D printing design and fabrication .

3. Root Cause Mapping: V1 Failures to V2 Solutions

The following table maps the failures experienced during the V1 iteration to the objective corrective actions implemented in the V2 sprint.

Failure Category	V1 Observation / Issue	Root Cause	V2 Corrective Action
Hardware / Sensor	Initial target of two side-mounted IR sensors to drive bidirectional motor movement failed. Both sensors triggered simultaneously.	High ambient IR noise and signal bounce within the testing room caused cross-talk.	**Pivot & Redesign:** Added 3D-printed sheaths/coverings to restrict IR sensor Field of View (FOV) to ~65 degrees. Also sink sensors into main body for a smooth outer body.

Failure Category	V1 Observation / Issue	Root Cause	V2 Corrective Action
Process / QA	Discovered the IR FOV overlap too late to implement a mechanical fix for the V1 deadline.	Testing of sensor angles and environmental reflection was scheduled too late in the sprint.	**Process Shift:** Objective, isolated hardware validation moved to the very beginning of the sprint backlog (Shift-Left Testing).
Deliverable	Unable to deliver the fully working targeting model for V1.	Hardware cross-talk prevented reliable closed-loop control.	**V1 Pivot:** Delivered a subsystem interaction demo (1 IR sensor → Arduino → simultaneous laser/motor activation).

4. Objective Tracking Methodology

To strictly adhere to the brief ("without relying on subjective tracking methods"), this sprint utilizes **Binary Task Completion** and **Test-Driven Acceptance**.

- **No Percentages:** A task is either `0%` (To Do/In Progress) or `100%` (Done). "90% done" is not recognized.
- **Decoupled Verification:** Implementation and testing are strictly separated into distinct backlog items to prevent developers from subjectively passing their own code without empirical validation.
- **Empirical Evidence:** A task is only marked `Done` when telemetry data, physical measurements, or test scripts prove the acceptance criteria have been met.

5. Product and Sprint Backlog

The following backlog represents the closed-loop fixes applied to V2 (currently marked as Done) and the prioritized To-Do items for the next iteration (Future).

5.1 Hardware sprint Backlog

5.1.1 Hardware Sprint Backlog (For V2)

Item ID	Title	Description	Assigned To	Status
HW-201	Install Additional IR Sensors	Wire and mount 5 additional IR sensors to the sensor ring.	Seth	Done
HW-202	Install 5V Regulator	Wire and integrate a 5V voltage regulator to stabilize power distribution.	Seth	Done
HW-203	Fabricate V2 Components	3D print new hanging mechanism, IR shrouds, IR ring, laser mount, and battery mount.	Allan	Done

Item ID	Title	Description	Assigned To	Status
HW-204	Assembly of V2 Parts	Assemble the new 3D printed mounts and attach them to the primary PVC chassis.	Allan	Done
HW-205	Test: IR Shroud FOV	Objectively verify that the new IR shrouds successfully restrict the sensor angle to ~65 degrees.	Allan	Done
HW-206	Test: Hanging Mechanism	Validate that the new hanging mechanism supports the chassis weight on the swivel without structural failure.	Seth	Done

5.1.2 Hardware Product Backlog (For V3)

Item ID	Title	Description	Assigned To	Status
HW-301	Install Power Switch	Wire a master power switch into the main battery circuit.	Seth	To Do
HW-302	Install Status LED	Wire an LED status light to indicate when the system is powered and active.	Seth	To Do
HW-303	Cable Management	Solder all cabling neatly to prevent wire snags inside the 150mm diameter PVC pipe constraints.	Allan	To Do
HW-304	Evaluate Analogue IR	Research and test swapping current sensors to analogue photodiode IR sensors for higher fidelity data.	Seth	To Do
HW-305	Install Piezo Speaker	Wire a piezoelectric speaker for audible status/error outputs.	Seth	To Do

Item ID	Title	Description	Assigned To	Status
HW-306	Fabricate Electronics Mount	3D print a dedicated mount to secure the Arduino and PCBs.	Allan	To Do
HW-307	Redesign Battery Mount	Redesign the V2 battery mount for better weight distribution (crucial for reaction wheel balance).	Allan	To Do
HW-308	Sensor positioning and chassis cutting	Cut out holes to mount shrouds and sensors internally to the bot.	Allan	To Do

5.2.1 Software Sprint Backlog (For V2)

Item ID	Title	Description	Assigned To	Status
SW-201	Implement New Sensor Logic	Add the necessary code to read data from the newly expanded array of 6 total IR sensors.	Harvey	Done
SW-202	Test: Sensor Array Verification	Objectively verify that the Arduino correctly reads independent signals from all sensors without crossover.	Harvey, Gabriel, Richard	Done
SW-203	Implement Control Logic States	Design and code the control logic framework to handle the various states (e.g., scanning, targeting, firing) the system will enter.	Gabriel	Done
SW-204	Implement Telemetry Scripts	Create Python scripts designed to generate charts and figures from raw telemetry and testing data.	Richard, Gabriel	Done
SW-205	Implement Data Gathering	Add the necessary Arduino and Python scripts to log data generated during hardware bench tests.	Gabriel, Harvey	Done

Item ID	Title	Description	Assigned To	Status
SW-206	Test: Motor Calibration	Test motor movement using test scripts to achieve precise movement calibration for the reaction wheel.	Richard, Gabriel	Done
SW-304	Implement V2 Targeting Process	Design and implement the full targeting process tailored to the new 360-degree sensor ring setup.	Gabriel, Harvey	Done
SW-301	Implement Targeting Algorithm	Design and write the script that calculates the required reaction wheel movement to orient the laser at a detected 38kHz beacon.	Gabriel	Done

5.2.2 Software Product Backlog (For V3)

Item ID	Title	Description	Assigned To	Status
SW-308	Implement V3 Targeting Process	Design and implement the full targeting process tailored to the analogue sensors.	Gabriel	Done
SW-304	Implement Improved Targeting Algorithm	Refactor original script to provide smoother movement and better precision and accuracy.	Gabriel, Harvey	Done
SW-302	Test: Scheduling Configs	Compare various software configurations for performance, specifically evaluating task scheduling methods for efficiency.	Richard	To Do

Item ID	Title	Description	Assigned To	Status
SW-303	Implement Spectrum Filtering	Add software spectrum filtering for the new sensors to further isolate the 38 kHz modulated light from ambient noise.	Harvey	To Do
SW-305	Full System Integration	Integrate all hardware devices through the Arduino control logic code to complete the full autonomous targeting task.	Gabriel, Harvey, Richard	To Do
SW-306	Software Debouncing Logic	Add debouncing algorithms to the IR sensor inputs to prevent false positive triggers from brief ambient light spikes.	Gabriel	To Do
SW-307	Hardware Interrupt Integration	Refactor the sensor reading logic to use hardware interrupts instead of constant polling to improve reaction time.	Harvey, Gabriel	To Do

5.3 Sprint Progress & Burndown Chart

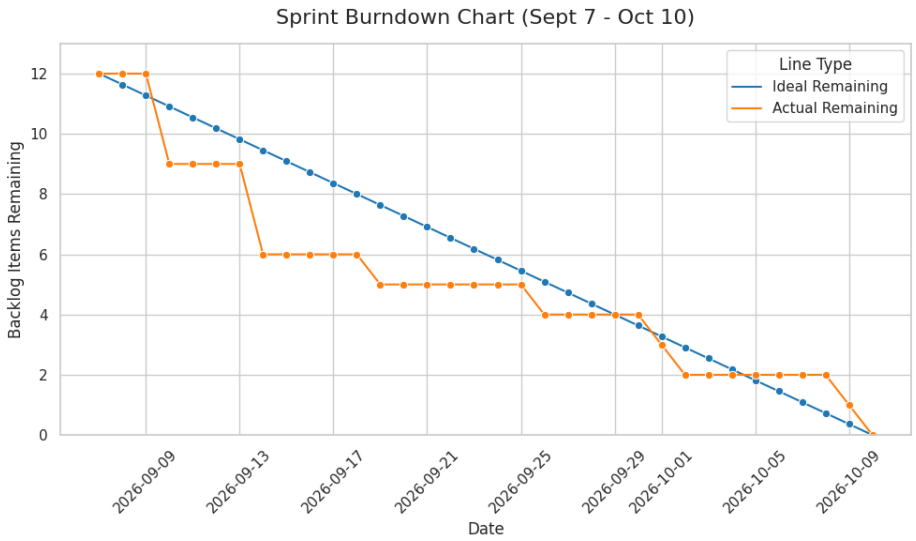


Figure 1: Burndown Chart

6. Definition of Done (DoD)

To maintain impeccable standard-paced execution, a User Story or Task is only considered "Done" when it meets the following criteria:

1. **Constraint Compliant:** The addition does not violate the 170mm x 150mm PVC pipe dimension constraints or utilize external propulsion.
2. **Code Quality:** Code compiles without warnings, contains no blocking `delay()` functions (unless strictly required for initialization), and is pushed to the main repository.
3. **Empirically Tested:** Objective telemetry or physical measurement confirms the feature works as intended (e.g., FOV is mathematically confirmed at $\leq 65°$).
4. **Peer Reviewed:** Software merges must be reviewed by at least one other software team member (Harvey, Richard, or Gabriel).